BIOST 2155: Assignment 3

YOUR NAME HERE (YOUR PITT EMAIL HERE)

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Instructions

- This file serves as a template for your assignment. Remember to set eval = TRUE below when compiling the report.

```
pacman::p_load(formatR)
knitr::opts_chunk$set(echo = TRUE, eval = FALSE,
                      fig.height = 4, fig.width = 6, fig.align = "center",
                      message=FALSE, tidy.opts = list(width.cutoff = 70), tidy = TRUE,
                      class.output  = "shadebox",
                      class.message = "shadebox",
                      class.warning = "shadebox",
                      class.error   = "shadebox")
```

- Use comments to annotate and explain your code whenever possible.
- This assignment uses one data set: `Wisconsin Breast Cancer data` and is saved as `assign3_Data.csv`

Theory (10 points)

In decision trees for classification, we need to select an impurity function to determine the best split to construct the tree. Gini index and entropy are two most common choices. Show that the Gini index, defined as $G = 1 - \sum_{k=1}^{K} p_k^2 = \sum_{k=1}^{K} p_k(1 - p_k)$ takes the maximum value when $p_1 = p_2 = \ldots = p_K = 1/K$ (the most impure). For simplicity, you may investigate with $K = 2$ ($p_1 = p$, $p_2 = 1 - p$) and express the indices in terms of just p. For full credit, your proof must be for general $K \geq 2$.

Hint: For $K \geq 2$, note that we have the constraint $\sum_{k=1}^{K} p_k = 1$. In order to find maximum of G with the given constraint, you may consider using the Lagrangian multiplier method for a calculus-based proof, or the Cauchy-Schwarz inequality for an algebraic proof.

Application (5 + 35 points = 40 points)

You are tasked with analyzing a classical dataset on breast cancer. The dataset contains `n = 569` observations and 31 features. We will be using 10 features for this problem. Features are computed from a digitized image of a fine needle aspirate (FNA) of a breast mass. They describe characteristics of the cell nuclei present in the image. The target labels (`M: malignant`/`B: benign`) indicate that the person's tumor is malignant (cancerous) or benign (noncancerous). Your goal is to explore this data, build predictive models to classify tumor type, and investigate the underlying structure of the data. **Please set a random seed (`set.seed(42)`) before any step involving data splitting or randomization to ensure your results are reproducible.**

```r
rm(list = ls())
pacman::p_load(tidyverse, caret, rpart, rpart.plot, randomForest, gbm,
    pROC, ggsci, here, tableone)
data <- read_csv(file.path(here(), "assign3_data.csv")) |>
    dplyr::select(-1)
```

Question 1: Initial Data Exploration and Preparation (4 + 1 = 5 Points)

Part A (4 points) Before building any models, it's crucial to understand the data. Create a table 1 that summarizes the ten features, stratified by `diagnosis`. You may use the `tableone` package for this. Please report and interpret mean ± standard deviation for all continuous variables and a column for the overall p-value to test for differences between the two groups.

```r
### table1 ###
```

Part B (1 points) Split the data into a training set (80%) and a test set (20%).

```r
### split into train and test ###
```

Question 2: PCA, PCR, Random Forest, and Ensemble building (2 + 5 + 5 + 5 + 10 + 5 + 3 = 35 points)

Part A (2 points) Explain why scaling (standardizing) features is a critical preprocessing step before applying Principal Component Analysis (PCA). In the space below, add code to standardize the ten features in both training and test sets.

```r
### centering and scaling ###
```

Part B (5 points) Perform PCA on the standardized training data. Plot two graphs: 1. The Proportion of Variance Explained (PVE) by each component (a "scree plot"). 2. The Cumulative PVE. Looking at your Cumulative PVE plot, how many components (`M`) are required to explain at least 75% of the variance?

```r
### scree (ew) plots ###
```

Part C (5 points) Create a scatterplot showing the first two principal component (PC1 vs. PC2) scores for your training data. Color the data points based on their diagnosis (`Malignant` or `Benign`). Do these first two components alone appear to provide good separation between the two classes?

```r
### scatterplot ###
```

Part D (5 points) Create a biplot showing the first two principal component (PC1 vs. PC2) scores for your training data. Examine the loading vectors (the arrows). Which original features contribute most heavily to PC1? Which contribute most to PC2?

```r
## biplot here ###
```

Part E (10 points) Using the first two principal components, create your new feature set for both the training and test sets. Fit a Logistic Regression model using the first two PCs as your predictors. Use `caret::confusionMatrix` to report model performance on the test set.

```
### PCR ###
```

Part F (5 points) Fit a `RandomForestClassifier` on the training data with `n.trees = 100`. Use `caret::confusionMatrix` to report model performance on the test set. Plot the variable importances from the trained model and list the top 5 important features.

```
### random forest ###
```

Part G (3 points) We will consider two models for malignant/benign classification: the PCR model and the random forest model, with the intention of making an ensemble model. Consider a 2-by-2 table with test set predictions from both models. Is there any disagreement between the two models? In case of disagreement, which model would you trust more?

```
### agreement between PCR and RF ###
```

Hint: go back to the confusion matrices from both models and check specificity and sensitivity. Since we want to minimize false negatives (instances where a patient with a malignant tumor was falsely classified as having a benign tumor), which metric should we consider when comparing the two models?